A FORMAL REDUCTION ARCHITECTURE FOR THE COLLATZ OPERATOR: 2-ADIC BLOCK DYNAMICS, DIOPHANTINE INTERFACES, AND FORMAL VERIFICATION

THOMAS HOFFBAUER

Abstract. We develop a conditional reduction architecture for the Collatz problem based on the accelerated map on positive odd integers. We derive the canonical block identity

$$2^{K_J(n)}C^J(n) = 3^J n + B_J(n)$$

and the exact subtraction-free criterion

$$3^J n + B_J(n) < 2^{K_J(n)} n$$

for strict block descent. The odd state space is partitioned according to the existence of a finite strong-contraction crossing time. Two explicitly open arithmetic interfaces are then isolated: a suffix-accumulation barrier in the crossing sector and deterministic integrality rigidity in its complement. Conditional on these interfaces, every positive orbit reaches a strictly smaller positive state. The abstract well-founded theorem that positive local descent implies global reachability of 1 is formalized in Lean 4 without additional axioms. No proof of either open arithmetic interface, and hence no proof of the Collatz conjecture, is claimed.

1. Introduction and scope

The $3x + 1$ problem resists elementary resolution because the 2-adic valuation sequence and Archimedean orbit size interact through a nonlocal affine term; see [2]. Tao proved that almost all Collatz orbits, in logarithmic density, attain almost bounded values [3]. Such generic decay does not imply the absence of exceptional positive-integer orbits.

This paper does not claim a proof of the Collatz conjecture and does not infer pointwise nonexistence from a measure-zero statement. Its purpose is to give an exact conditional factorization. The order-theoretic component—positive local descent implies global reachability of 1—is mechanized in Lean 4. The unresolved arithmetic is isolated in two target interfaces:

(1) *Deterministic integrality rigidity* (IR): an arithmetic obstruction preventing an exceptional formal 2-adic fiber from being realized by a positive integer;
(2) *Uniform suffix barrier* (SO): a quantitative obstruction to persistent affine overcompensation in reverse valuation-excess sequences.

The labels IR and SO denote open targets, not established theorems. Their usefulness depends on whether they admit independent formulations and proofs that do not already assume finite descent.

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Component	Mathematical content	Status
Reaches$(T,\cdot,\cdot)$	Reflexive-transitive closure and transitivity	Lean 4 verified; no axioms
`collatz_well_founded_closure`	Strong induction on $(\mathbb{N},<)$	Lean 4 verified; no axioms
`even_reaches_smaller_odd`	Descent from a positive even integer	Lean 4 verified; no axioms
`master_conditional_reduction`	Conditional bridge from local descent to 1	Lean 4 verified; no axioms
Canonical block identity	$2^{K_J}C^J(n) = 3^J n + B_J(n)$	Proved algebraically here
Accelerated reachability	Reaches$(T, n, C^J(n))$	Proved by induction here
Spectral radius $\rho_\infty < 1$	Proposed transfer-operator contraction	Numerical motivation only
Integrality rigidity IR	Exclusion of positive-integer exceptional fibers	Open interface
Suffix barrier SO	Uniform obstruction to affine overcompensation	Open interface

TABLE 1. Epistemic and verification status of the framework components.

2. Operators, reachability, and block dynamics

Let $\mathbb{N} = \{0, 1, 2, \ldots\}$ and

$$\mathcal{O} = \{n \in \mathbb{N} : n \geq 1 \text{ and } 2 \nmid n\}.$$

Define $T : \mathbb{N} \to \mathbb{N}$ by

$$T(n) = \begin{cases} 0, & n = 0, \\ 1, & n = 1, \\ n/2, & n \geq 2 \text{ even}, \\ 3n+1, & n \geq 3 \text{ odd}. \end{cases} \tag{1}$$

For a map $S : \mathbb{N} \to \mathbb{N}$, write

$$\text{Reaches}(S, x, y) \quad \Longleftrightarrow \quad \exists r \in \mathbb{N} : \ S^r(x) = y. \tag{2}$$

Thus Reaches is reflexive and transitive. The accelerated odd map $C : \mathcal{O} \to \mathcal{O}$ is

$$C(n) = \frac{3n+1}{2^{\nu_2(3n+1)}}. \tag{3}$$

For $n \in \mathcal{O}$ and $i \geq 0$, put

$$a_i(n) = \nu_2\big(3C^i(n)+1\big), \qquad K_J(n) = \sum_{i=0}^{J-1} a_i(n), \qquad A_i(n) = \sum_{r=0}^{i-1} a_r(n), \tag{4}$$

where $A_0(n) = 0$.

Proposition 2.1 (Canonical block identity)**.** *For $n \in \mathcal{O}$ and $J \geq 1$,*

$$2^{K_J(n)} C^J(n) = 3^J n + B_J(n), \qquad B_J(n) = \sum_{i=0}^{J-1} 3^{J-1-i} 2^{A_i(n)}. \tag{5}$$

Proof. For $J = 1$, the defining relation $2^{a_0(n)}C(n) = 3n + 1$ gives the result. Assume the identity at J. Since $2^{a_J(n)}C^{J+1}(n) = 3C^J(n) + 1$, multiplication by $2^{K_J(n)}$ gives

$$2^{K_{J+1}(n)}C^{J+1}(n) = 3\big(3^J n + B_J(n)\big) + 2^{K_J(n)}.$$

The right-hand side is $3^{J+1}n + B_{J+1}(n)$ by the displayed formula for B_J. This proves the claim by induction. □

Corollary 2.2 (Subtraction-free integer descent)**.** *For $n \in \mathcal{O}$, $n > 1$, and $J \geq 1$,*

$$C^J(n) < n \quad \Longleftrightarrow \quad 3^J n + B_J(n) < 2^{K_J(n)} n. \tag{6}$$

Proof. Substitute (5) and use $2^{K_J(n)} > 0$. □

Lemma 2.3 (Accelerated-to-classical reachability)**.** *For $n \in \mathcal{O}$ and $J \geq 0$,*

$$\mathrm{Reaches}\big(T, n, C^J(n)\big).$$

Proof. If the current odd state is 1, both maps remain at 1. If it is greater than 1, one application of T produces $3n + 1$, followed by exactly $\nu_2(3n + 1)$ divisions by 2, and reaches $C(n)$. Induction on J and transitivity complete the proof. □

3. Crossing partition and exact suffix algebra

Define the logarithmic discrepancy

$$D_m(n) = K_m(n) \log 2 - m \log 3. \tag{7}$$

Definition 3.1 (Crossing partition)**.** For $n \in \mathcal{O}$, $n > 1$, define

$$\mathrm{SectorB}(n) \Longleftrightarrow \exists m \geq 1 : \ 2^{K_m(n)} \geq 2 \cdot 3^m, \tag{8}$$

$$\mathrm{SectorA}(n) \Longleftrightarrow \forall m \geq 1 : \ 2^{K_m(n)} < 2 \cdot 3^m. \tag{9}$$

In Sector B let

$$\tau_B(n) = \min\{m \geq 1 : 2^{K_m(n)} \geq 2 \cdot 3^m\}.$$

These predicates are disjoint and exhaustive. Equivalently, Sector B has $D_m(n) \geq \log 2$ for some m, while Sector A has $D_m(n) < \log 2$ for every $m \geq 1$.

For a block of length M, define the suffix valuation and suffix excess

$$D_{j,M}(n) = \sum_{\ell=j}^{M-1} a_\ell(n), \tag{10}$$

$$G_{j,M}(n) = D_{j,M}(n) - (M - j) \log_2 3, \qquad 0 \leq j < M, \tag{11}$$

and put $q_M(n) = 3^M / 2^{K_M(n)}$. Since $D_{j,M} = K_M - A_j$, Proposition 2.1 gives the exact identity

$$\frac{C^M(n)}{n} = q_M(n) + \frac{1}{3n} \sum_{j=0}^{M-1} 2^{-G_{j,M}(n)}, \tag{12}$$

$$1 - \frac{C^M(n)}{n} = \big(1 - q_M(n)\big) - \frac{1}{3n} \sum_{j=0}^{M-1} 2^{-G_{j,M}(n)}. \tag{13}$$

The coefficient $1/3$ is forced by the exponent $M-j$ in the definition of $G_{j,M}$. Alternatively, it can be absorbed by defining the excess with $M-1-j$ instead.

The negative suffix mass

$$\mathcal{N}_M(n) = \sum_{\substack{0\le j<M \\ G_{j,M}(n)<0}} 2^{|G_{j,M}(n)|} \tag{14}$$

isolates suffixes that enlarge the affine contribution. No claim is made that descent alone forces $\mathcal{N}_M(n)=0$; such a statement would require an additional theorem.

Remark 3.2 (A first-crossing estimate)**.** Let $J=\tau_B(n)$. Minimality gives $2^{K_i(n)} < 2\cdot 3^i$ for $i<J$. Hence

$$B_J(n) < 2J3^{J-1}.$$

Moreover $2^{K_J(n)} - 3^J \ge 3^J$. Consequently, $2J<3n$ is a sufficient, not necessary, condition for $C^J(n)<n$. A universal estimate ensuring this condition is not established here.

4. Open arithmetic interfaces

The following hypotheses state the unresolved content needed by the conditional reduction. In their present form they are target interfaces. Because their conclusions are descent inequalities, they must not be counted as independent progress until replaced by, or derived from, genuinely weaker and independently verifiable arithmetic conditions.

Hypothesis 4.1 (Sector-B suffix barrier (SO))**.** *For every $n\in\mathcal{O}$, $n>1$, satisfying* SectorB(n)*, there exists $M\ge 1$ such that the suffix contribution obeys*

$$\frac{1}{3n}\sum_{j=0}^{M-1} 2^{-G_{j,M}(n)} < 1 - q_M(n). \tag{15}$$

Equivalently, $C^M(n)<n$ by (13).

Hypothesis 4.1 deliberately displays the precise affine obstruction. A substantive suffix theorem would supply an independently stated bound on the sum in (15), for example through constraints on the negative suffix mass, and would then imply this interface.

Hypothesis 4.2 (Sector-A deterministic integrality rigidity (IR))**.** *No positive integer $n>1$ can both satisfy* SectorA(n) *and avoid finite descent under C. Equivalently, for every such n there exists $J\ge 1$ with*

$$3^J n + B_J(n) < 2^{K_J(n)} n. \tag{16}$$

The intended research program for Hypothesis 4.2 is to derive it from an independently formulated incompatibility between positive-integer realizability, the inverse congruence system of a valuation word, and persistent Sector-A behavior. Neither a numerical spectral radius below 1 nor a measure-zero exceptional set suffices: a measure-zero set may still contain positive integers. Bounds for nonzero linear forms $K\log 2 - m\log 3$ [1] control proximity to resonance but do not by themselves force a crossing or a descent.

5. Well-founded closure

Theorem 5.1 (Well-founded positive closure)**.** *Let $S : \mathbb{N} \to \mathbb{N}$. Suppose every $n > 1$ reaches a state y with $1 \le y < n$. Then every $n \ge 1$ reaches 1. If additionally $S(1) = 1$, then 1 is absorbing.*

Proof. Use strong induction on n. The case $n = 1$ follows by reflexivity. For $n > 1$, choose $1 \le y < n$ reachable from n. By the induction hypothesis, 1 is reachable from y, and transitivity completes the proof. The corresponding Lean 4 theorem uses `Nat.strong_induction_on`. □

6. Conditional master reduction

Theorem 6.1 (Conditional master reduction)**.** *Assume Hypotheses 4.1 and 4.2. Then every positive integer reaches 1 under T.*

Proof. It is enough, by Theorem 5.1, to obtain a smaller positive reachable state from every $n > 1$. If n is even, write $n = 2^r y$ with $r \ge 1$ and y odd. Repeated division by 2 reaches y, where $1 \le y < n$.

If n is odd, the crossing partition places it in exactly one sector. Hypothesis 4.1 in Sector B or Hypothesis 4.2 in Sector A, together with Corollary 2.2, gives an $M \ge 1$ for which $y = C^M(n) < n$. The value y is positive, and Lemma 2.3 transfers the accelerated block to T-reachability. Theorem 5.1 now applies. □

7. Formal-verification boundary

The accompanying Lean development verifies the reachability infrastructure, the well-founded closure, the even-state reduction, and the abstract conditional bridge. The exact status of each theorem is recorded in Table 1. In particular, the numerical observation of a putative subcritical transfer-operator spectral radius $\rho_\infty \approx 0.9659$ is empirical motivation only. It is neither used as a theorem nor interpreted as excluding individual integer points.

The block and suffix identities are proved on paper above. Unless and until their concrete Lean counterparts are included in the checked source, they are not described as machine verified. Hypotheses 4.1 and 4.2 remain open and contain the unresolved arithmetic content.

8. Conclusion

The conditional architecture is summarized by

$$\boxed{\text{IR}} \wedge \boxed{\text{SO}} \implies \boxed{\text{positive local descent}}$$
$$\implies \boxed{\text{global reachability of 1}}.$$

The final implication is elementary and machine checked. The first implication is conditional on two open interfaces. In their current target form, those interfaces should not be mistaken for solved subproblems or for an unconditional reduction in difficulty. The framework's contribution is the explicit separation of exact block algebra, empirical motivation, open arithmetic input, and formal order-theoretic closure.

References

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Bamberg, Germany
Email address: t.hoffbauer@cibamberg.de